

Installation Handbook

This handbook describes how Northwind Solar crews install rooftop systems. Every job starts with a structural survey of the roof, carried out by a certified technician from the Bergen workshop or the Rotterdam office.

The survey checks rafter spacing, roof age and load capacity. A roof older than 25 years must be inspected by a structural engineer before panels are mounted.

Shading analysis

A shading analysis estimates the annual energy yield from drone images and local weather data. If any panel position is shaded for more than two hours a day, the crew fits Tidewater Micro microinverters instead of a single Tidewater string inverter.

Mounting

NW-400 panels are mounted on anodised aluminium rails bolted into the roof rafters. Every bolt is sealed with butyl tape to keep water out. On coastal jobs all fasteners are stainless steel, because salt spray is the main cause of corrosion.

Rails are spaced 1.2 metres apart. The crew torques each bolt to 25 newton-metres and records the value in the job sheet.

Electrical work

Inverters from Kestrel Power Systems are installed in a shaded, ventilated location. The Fjord LFP-10 battery, when ordered, goes in a ventilated cabinet away from direct sunlight, because high temperatures shorten battery life.

Commissioning ends with a two-hour monitored test run. The inverter logs are uploaded to the customer portal before the crew leaves the site.

Annual maintenance

Technicians visit once a year. They clean salt deposits from the panel glass, check that the rail bolts are still at 25 newton-metres, and download inverter logs to find underperforming strings.